Chapter 9: Circles and Tangents

In the preceding chapters we developed the notions of betweenness, segment congruence, linear objects (segments, rays, lines), and the separation of the plane. To proceed to the theory of circles, perpendicularity, and tangents, the elementary axioms introduced so far are not sufficient. We must extend our formal system with additional axioms that govern the transport of segments, the addition of congruent segments, and the existence of perpendicular lines. We state these required axioms explicitly before proving any further theorems. All subsequent reasoning will rest solely on the primitive notions and the axioms (both old and new).

1 Summary of Previously Required Axioms

For completeness, we restate the axioms from earlier chapters that are used in the proofs below. Their original numbering is preserved so that `\ref` calls in later proofs will resolve correctly.

Axiom 1.1 (Symmetry of Betweenness). $\forall A, B, C \in \mathcal{P},\ \mathsf{B}(A, B, C) \implies \mathsf{B}(C, B, A)$.

Axiom 1.2 (Identity of Betweenness). $\forall A, B \in \mathcal{P},\ \mathsf{B}(A, B, A) \implies A = B$.

Axiom 1.3 (Extensibility). $\forall A, B \in \mathcal{P},\ A \neq B \implies \exists C \in \mathcal{P} : \mathsf{B}(A, B, C)$.

Axiom 1.4 (Density). $\forall A, C \in \mathcal{P},\ A \neq C \implies \exists B \in \mathcal{P} : \mathsf{B}(A, B, C)$.

Axiom 1.5 (Transitivity of Betweenness). $\mathsf{B}(A, B, C) \wedge \mathsf{B}(B, C, D) \wedge B \neq C \implies \mathsf{B}(A, B, D)$.

Axiom 1.6 (Reflexivity of Congruence). $\forall A, B \in \mathcal{P},\ AB \cong AB$.

Axiom 1.7 (Reversal of Congruence). $\forall A, B \in \mathcal{P},\ AB \cong BA$.

Axiom 1.8 (Transitivity of Congruence). $\forall A, B, C, D, E, F \in \mathcal{P},\ (AB \cong CD \wedge AB \cong EF) \implies CD \cong EF$.

Axiom 1.9 (Midpoint Existence). $\forall A, B \in \mathcal{P},\ A \neq B \implies \exists M \in \mathcal{P} : (\mathsf{B}(A, M, B)) \wedge (AM \cong MB)$.

Axiom 1.10 (Plane Separation). *For every line $\ell \in \mathcal{L}$, there exist two sets $H_{\ell,1}, H_{\ell,2} \subset \mathcal{P} \setminus \ell$ such that:*

1. $H_{\ell,1} \cup H_{\ell,2} = \mathcal{P} \setminus \ell \ \wedge\ H_{\ell,1} \cap H_{\ell,2} = \emptyset$;
2. $\forall A, B \in \mathcal{P} \setminus \ell :\ \big(\exists X \in \ell,\ \mathsf{B}(A, X, B)\big) \iff \{A, B\} \not\subset H_{\ell,1} \wedge \{A, B\} \not\subset H_{\ell,2}$.

2 Additional Axioms Required for Circles

Axiom 2.1 (Segment Transport)**.** *Let $A, B, C, D \in \mathcal{P}$ with $C \neq D$. Let $\overrightarrow{CD}$ be the ray from C through D. There exists a unique point $E \in \overrightarrow{CD}$ such that $CE \cong AB$.*

Axiom 2.2 (Segment Addition)**.** *If $\mathsf{B}(A, B, C)$ and $\mathsf{B}(D, E, F)$, and if $AB \cong DE$ and $BC \cong EF$, then $AC \cong DF$.*

Axiom 2.3 (Perpendicular Bisector)**.** *For any two distinct points $A, B \in \mathcal{P}$ the set*

$$\ell = \{\, X \in \mathcal{P} \mid AX \cong BX \,\}$$

is a line. It is called the perpendicular bisector *of AB. The line ℓ intersects the line $\overleftrightarrow{AB}$ at the midpoint of segment AB.*

Axiom 2.4 (Existence of the Perpendicular)**.** *Let ℓ be a line and $O \notin \ell$ a point. There exists a unique point $F \in \ell$ such that the line $\overleftrightarrow{OF}$ is perpendicular to ℓ (see Definition 2.1 below). The point F is called the* foot *of the perpendicular from O to ℓ.*

2.1 Perpendicular Lines

Definition 2.1 (Perpendicular Lines)**.** *Two lines l and m are* perpendicular*, denoted by $l \perp m$, if they intersect at a point O and there exist points $A, B \in m$ such that O is the midpoint of AB and $l = \{\, X \in \mathcal{P} \mid AX \cong BX \,\}$. In other words, l is the perpendicular bisector of a segment that lies on m and has O as its midpoint.*

From Axioms 2.3 and 2.4 together with the definition it follows that for any segment AB the perpendicular bisector is indeed perpendicular to the line $\overleftrightarrow{AB}$. Also, through any point O not on a line ℓ there is exactly one line through O perpendicular to ℓ.

3 Circles

Definition 3.1 (Circle)**.** *Let $O \in \mathcal{P}$ be a point and let OR be a segment. The* circle *with center O and radius segment OR is the set*

$$\mathcal{C}(O, OR) \;:=\; \{\, P \in \mathcal{P} \mid OP \cong OR \,\}.$$

A point P is said to lie on the circle *if $P \in \mathcal{C}(O, OR)$.*

Definition 3.2 (Chord, Diameter, Radius)**.** *Let $\mathcal{C}(O, OR)$ be a circle.*

- *A segment AB is a* chord *of the circle if both A and B lie on the circle and $A \neq B$.*
- *A chord AB is a* diameter *if the center O lies on the segment AB (i.e. $\mathsf{B}(A, O, B)$).*
- *Any segment joining the center O with a point on the circle is called a* radius segment*. The segment OR that defines the circle is one such radius segment; by transitivity all radius segments of the same circle are congruent to one another.*

4 Properties of Chords

Theorem 4.1 (Center to Midpoint of a Chord)**.** *If a line through the center O of a circle $\mathcal{C}(O, OR)$ passes through the midpoint M of a chord AB, then that line is perpendicular to the chord.*

Proof. Because A and B lie on the circle, we have $OA \cong OR$ and $OB \cong OR$; by symmetry and transitivity of congruence (Axiom 1.8), $OA \cong OB$. Hence O belongs to the set $\{ X \in \mathcal{P} \mid AX \cong BX \}$. By Axiom 2.3 this set is the perpendicular bisector of segment AB; call it ℓ. The line ℓ contains both the midpoint M of AB and the center O. Therefore $\ell = \overleftrightarrow{OM}$ is, by definition, perpendicular to $\overleftrightarrow{AB}$. Thus the line through the center and the midpoint is perpendicular to the chord. □

Theorem 4.2 (Perpendicular from Center to Chord Bisects It)**.** *The line through the center O of a circle perpendicular to a chord AB bisects the chord.*

Proof. Let m be the line through O perpendicular to $\overleftrightarrow{AB}$. Because A and B lie on the circle, $OA \cong OB$, so O lies on the perpendicular bisector of AB. By Axiom 2.3 the perpendicular bisector of AB is unique and is a line perpendicular to $\overleftrightarrow{AB}$ passing through the midpoint of AB. The line m also passes through O and is perpendicular to $\overleftrightarrow{AB}$; therefore m must coincide with the perpendicular bisector of AB. Consequently m contains the midpoint of AB, i.e. the perpendicular from the center bisects the chord. □

5 Tangents

Definition 5.1 (Tangent Line)**.** *A line ℓ is* tangent *to a circle $\mathcal{C}(O, OR)$ if ℓ and the circle have exactly one point in common:*

$$\ell \cap \mathcal{C}(O, OR) = \{T\}.$$

The point T is called the point of tangency*.*

Theorem 5.1 (Radius Perpendicular to Tangent)**.** *If a line ℓ is tangent to the circle $\mathcal{C}(O, OR)$ at a point T, then the radius segment OT is perpendicular to ℓ: $\overleftrightarrow{OT} \perp \ell$.*

Proof. We argue by contradiction. Suppose ℓ is tangent at T but $\overleftrightarrow{OT}$ is *not* perpendicular to ℓ. The center O cannot lie on ℓ because T is on the circle, so $OT \cong OR$ while OO (degenerate) is not congruent to OR. Hence $O \notin \ell$.

Apply Axiom 2.4 (Existence of the Perpendicular) to the line ℓ and the point O. There exists a unique foot $F \in \ell$ such that $\overleftrightarrow{OF} \perp \ell$. Since $\overleftrightarrow{OT}$ is not perpendicular to ℓ, the point F is different from T.

The line $\overleftrightarrow{OF}$ is perpendicular to ℓ; by Definition 2.1 this means that $\overleftrightarrow{OF}$ is the perpendicular bisector of some segment on ℓ whose midpoint is F. In particular, if we take the segment on ℓ whose endpoints are the two points obtained by laying off FT on both sides of F, we can construct a point T' on the ray opposite to $\overrightarrow{FT}$ such that $FT' \cong FT$ (using

Segment Transport, Axiom 2.1). Then F is the midpoint of TT' and, because $\overleftrightarrow{OF}$ is the perpendicular bisector of TT', every point on $\overleftrightarrow{OF}$ is equidistant from T and T'; in particular,

$$OT \cong OT'.$$

(The explicit verification uses the definition of perpendicular bisector: since $\overleftrightarrow{OF}$ is the perpendicular bisector of some segment with midpoint F on ℓ, and TT' lies on ℓ with midpoint F, the line $\overleftrightarrow{OF}$ is also the perpendicular bisector of TT'; the uniqueness of the perpendicular bisector follows from Axiom 2.3.)

Now, T lies on the circle, so $OT \cong OR$. By transitivity of congruence (Axiom 1.8), $OT' \cong OR$, which means $T' \in \mathcal{C}(O, OR)$. Moreover, T' lies on ℓ by construction. Because $F \neq T$ and $FT' \cong FT$, we have $T' \neq T$. Thus ℓ contains the two distinct points T and T' of the circle, contradicting the hypothesis that ℓ is tangent (intersects the circle in exactly one point).

The contradiction forces us to reject the assumption that $\overleftrightarrow{OT}$ is not perpendicular to ℓ. Hence $\overleftrightarrow{OT} \perp \ell$, as required. □

The converse statement—if a line through a point of the circle is perpendicular to the radius, then it is tangent—can be proved similarly using the uniqueness of the perpendicular and the fact that any other point on the line would violate the definition of the perpendicular bisector. We omit the proof here as it follows the same pattern.